

## teacups – Quick Reference

| attribute                 | content                                                                                                   | type                                          |
|---------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Sys.spin_system           | define spin system: “rp” (radical pair), “doub” (doublet), “trip” (triplet), “tdp” (triplet doublet pair) | string                                        |
| Sys.precursor             | define precursor state: “zf”, “eigen”, “singlet”, “triplet-zf”, “triplet-pnm”                             | string                                        |
| Sys.population            | populations of the initial density matrix diagonals                                                       | list of floats                                |
| Sys.g1                    | diagonal elements of the g1-tensor (rp-acceptor)                                                          | list of floats                                |
| Sys.g1_frame              | euler angles, rotate g1-tensor to the lab. frame                                                          | list of floats                                |
| Sys.g2                    | diagonal elements of the g2-tensor (rp-donor)                                                             | list of floats                                |
| Sys.g2_frame              | euler angles, rotate g2-tensor to the lab. frame                                                          | list of floats                                |
| Sys.g_tri                 | diagonal elements of the g-tensor of a triplet                                                            | list of floats                                |
| Sys.g_tri_frame           | euler angles, rotate g_tri to the lab. frame                                                              | list of floats                                |
| Sys.g                     | diagonal elements of the g-tensor of a doublet                                                            | list of floats                                |
| Sys.g_frame               | euler angles, rotate g to the lab. frame                                                                  | list of floats                                |
| Sys.width_gauss           | gaussian line width in mT                                                                                 | float                                         |
| Sys.donor_list            | index of cores assigned to the donor                                                                      | list of str/int                               |
| Sys.acceptor_list         | index of cores assigned to the acceptor                                                                   | list of str/int                               |
| Sys.A <sub>i</sub>        | diagonal elements of A <sub>i</sub> -tensor ( $1 \leq i \leq 5$ )                                         | list of floats                                |
| Sys.A <sub>i</sub> _frame | euler angles, rotate A <sub>i</sub> -tensor to the lab. frame                                             | list of floats                                |
| Sys.I <sub>i</sub>        | spin quantum number of core group <i>i</i> (integer or half)                                              | float                                         |
| Sys.n <sub>i</sub>        | number of cores in core group <i>i</i>                                                                    | integer                                       |
| Sys.A                     | diagonal elements of all A-tensors (assigned to spins <i>i</i> and optionally <i>j</i> )                  | list [[A <sub>i</sub> ], [A <sub>j</sub> ]]   |
| Sys.A_frame               | euler angles, rotate all A-tensors to the lab. frame                                                      | list [[Fr <sub>i</sub> ], [Fr <sub>j</sub> ]] |
| Sys.I                     | spin quantum numbers of all cores                                                                         | list [[A <sub>i</sub> ], [A <sub>j</sub> ]]   |
| Sys.D                     | axial dipolar coupling between two spins in MHz                                                           | float                                         |

| attribute            | content                                                                                           | type           |
|----------------------|---------------------------------------------------------------------------------------------------|----------------|
| Sys.E                | rhombic dipolar coupling between two spins in MHz                                                 | float          |
| Sys.J_ex             | exchange-coupling constant $J$ of two interacting electronic species $J$ in MHz                   | float          |
| Sys.D_tri            | zero-field-splitting constant $D$ of a triplet in MHz                                             | float          |
| Sys.E_tri            | zero-field-splitting constant $E$ of a triplet in MHz                                             | float          |
| Sys.D_frame          | euler angles, rotate D-tensor to lab. frame                                                       | list of floats |
| Sys.D_tri_frame      | euler angles, rotate triplet D-tensor to lab. frame                                               | list of floats |
| Sys.T_relax_1        | longitudinal relaxation time in s                                                                 | float          |
| Sys.T_relax_2        | transversal relaxation time in s                                                                  | float          |
| Sys.decay            | decay time (exponential) for hilbert space calculations in s                                      | float          |
| Sys.dynamics         | Matrix with rateconstants of dynamic processes in 1/s                                             | np.ndarray     |
| Exp.B_z              | magnetic field array in mT                                                                        | np.ndarray     |
| Exp.t_scale          | borders of the time variation in s                                                                | list of floats |
| Exp.t_points         | number of time field points                                                                       | integer        |
| Exp.B_mw             | amplitude of the microwave field in mT                                                            | float          |
| Exp.freq_mw          | microwave frequency in Hz                                                                         | float          |
| SimOpt.grid_points   | number of points between $\vartheta = 0$ and $\vartheta = \pi/2$ on the orientational sphere      | integer        |
| SimOpt.grid          | Type of grid on the sphere for powder averaging. Can be either "fibonacci" or "sophe" or "single" | string         |
| SimOpt.theta         | User-chosen orientation-angles for theta if SimOpt.grid = "single"                                | list           |
| SimOpt.phi           | User-chosen orientation-angles for phi if SimOpt.grid = "single"                                  | list           |
| SimOpt.sym           | Point group of the spin system, if SimOpt.grid = "sophe"                                          | string         |
| SimOpt.space         | define in which space the time evolution will be calculated, either "hilbert" or "liouville"      | string         |
| SimOpt.pop_evolution | Decides wether the population evolution is calculated and returned or not                         | boolean        |
| SimOpt.eigval_mode   | Decides wether only the eigenvalues shall be calculated                                           | boolean        |

| attribute        | content                                                                 | type    |
|------------------|-------------------------------------------------------------------------|---------|
| SimOpt.cpu_cores | Number of CPU cores used for time evolution, 0 uses all available cores | integer |
| SimOpt.CUPY      | Use GPU and Cupy for faster calculations                                | boolean |